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1. Use-Case Specification and Use-Case Models for Function 01: Authentication and Onboarding

1.1 Auth & Onboarding Use Case Diagram (Remember Login)

Authors: [Nguyễn Khánh Lin]h | **Reviewer:** [Minh Thư] | **Editor:** [Nguyễn Khánh Linh]

Changes

- Reversed the <<extend>> relationship direction between Remember Login and Login in the Module 1 use-case diagram, changing Login -.->|extend| Remember Login to Remember Login -.->|extend| Login.
- Removed the direct actor association Registered User --- Remember Login from the use-case diagram.

Reason for Change

- Per UML convention, the <<extend>> arrow must point from the extending use case to the base use case, since Remember Login is an optional behavior added to Login, not the reverse.
- Remember Login is only triggered inside the flow of Login (when the user checks the "Remember me" option) via the <<extend>> relationship, so it should not carry a direct actor association;

actors connect directly only to base use cases they independently initiate.

2. Use-Case Specification and Use-Case Models for Function 02: Quiz and Assessment

1.1 Module 2 Use Case Specifications

Authors: [Lê Kim Hằng] | **Reviewer:** [Minh Thư] | **Editor:** [Lê Kim Hằng]

Changes

- Resolved merge conflict in `UC5_spec.md` by removing duplicate 27-line incomplete version.
- Standardized author attribution across 7 UC files (UC3, UC4, UC8, UC12, UC13, UC14, UC15) from `Kim Hằng / System Analyst` to ``Lê Kim Hằng``.
- Created consolidated specification file `M2_specs.md` combining all 15 UC specifications (UC1–UC15) with comprehensive Table of Contents.
- Added 12 alternative flow diagrams to visual documentation across UC specifications.
- Fixed use case reference naming consistency: updated UC8 references from `"View Quiz Result"` to `"View Quiz/Assessment Result"` in UC5 and UC9.

Reason for Change

- Removed unresolved merge conflict that caused duplicate and conflicting content in Take Quiz use case specification.
- Standardized author identification to ensure consistent documentation practices and clear responsibility tracking.
- Consolidated 15 separate UC files into single document aligned with Module 4 and 5 patterns for improved maintainability and navigation.
- Enhanced visual documentation with alternative flow diagrams for complete specification coverage.
- Corrected incomplete use case naming references to improve traceability and prevent confusion in specification relationships.

3. Use-Case Specification and Use-Case Models for Function 03: Virtual Study Room

1.1 Project Overview

Authors: [Minh Thư] | **Reviewer:** [Thiên Phước] | **Editor:** [Minh Thư]

Changes

- Consolidated 7 individual use cases under Module03 into a single master document.
- Reviewed and updated all embedded images, media paths, and document links to ensure complete consistency and fix rendering/link errors.

Reason for Change

- Centralization & Maintenance: Streamlines documentation structure and improves readability by combining fragmented use cases into one unified file.
- Accuracy & Error Prevention: Ensures all visual assets and reference paths align correctly within the merged document structure to avoid broken links and missing image errors.

4. Use-Case Specification and Use-Case Models for Function 05: Flashcard

1.1 Module 4 Use Cases

Authors: [Thiên Phước] | **Reviewer:** [Minh Thư] | **Editor:** [Thiên Phước]

Changes

- Refactored Module 4 Use Case Model in `report.md` to remove invalid `<<include>>` workflows.
- Merged fragmented Use Case Specifications into 3 main Use Cases (`M4-UC1`, `M4-UC2`, `M4-UC3`).
- Added `M4-` prefix to all Module 4 Use Case IDs to ensure global uniqueness.
- Inserted missing UI Prototypes for Alternative Flows into the new specifications.

Reason for Change

- Address TA feedback regarding "abusing `<<include>>` as a workflow" (Functional Decomposition).
- Resolve duplicate Use Case IDs across modules.
- Ensure Use Case diagrams strictly adhere to UML standards.

5. Use-Case Specification and Use-Case Models for Function 05: AI Speaking Assistant

1.1 Project Overview

Authors: [Gia Phúc] | **Reviewer:** [Minh Thư] | **Editor:** [Gia Phúc]

Changes

- Refactored Module 5 Use Case Model to introduce UC5 as the parent Use Case and remove invalid `<<include>>` workflows.
- Updated UC5 to explicitly include UC5.1, UC5.2, and UC5.3, with UC5.4 extending UC5.3.
- Removed redundant direct `<<include>>` calls to UC5.1 from UC5.2 and UC5.3 specs.
- Merged all specs into one spec file

Reason for Change

- Resolve evaluator feedback on "abusing `<<include>>` as a workflow" (Functional Decomposition).
- Align written specifications strictly with standard UML Use Case rules.
- Resolve the problem of "UC specs should be merged into one file"